

Activity 15 Instruction Manual

Activity 15 | Topic 02 | C013 | Version v1.0

Purpose

This is the heart of LO2. A resource is over-allocated and the budget is under pressure - you will use GenAI to generate genuine alternatives, evaluate the trade-offs, and defend the option you choose.

Files used

- activity-guide.md and activity-guide.pdf - brief and source instructions
- prompts.pdf - reusable prompt and verification checklist
- mock-data.xlsx - safe fictional project data for practice

Expected output

A resource optimisation decision with a documented trade-off analysis.

Before you begin

1. Form the required group and assign one person to capture evidence.
2. Open mock-data.xlsx and review the title, note and column headings.
3. Open prompts.pdf beside your chosen GenAI tool. Do not upload confidential workplace data.

Detailed procedure

1. Take your schedule from Activity 9 and deliberately create a problem: assign one person to two activities running in the same period, and cut the budget by 15%.
2. Confirm the two techniques before prompting: resource LEVELLING adjusts the schedule to fix over-allocation and may move the critical path; resource SMOOTHING adjusts only within available float, so the critical path does not change.
3. Prompt with the Alternative Approaches pattern: 'I have an over-allocated resource on these activities [paste] and a 15% budget cut. Suggest at least three options - including resource levelling, resource smoothing, crashing and fast tracking. Compare pros, cons, cost impact and schedule impact in a table.'
4. Check each option against the definitions. GenAI frequently confuses levelling with smoothing and describes crashing without noting it only works on the critical path.
5. For any crashing option, verify it targets a critical path activity - crashing a non-critical activity buys you nothing. For fast tracking, identify the rework risk it creates.
6. Apply the Cognitive Verifier pattern: 'Break down which option gives the best schedule outcome for the least cost increase and the least added risk, then combine into a recommendation.'
7. Make your group's decision, adjust the Gantt chart accordingly, and write three sentences justifying it to a sponsor.
8. Present your recommendation and state one thing GenAI got wrong about levelling, smoothing, crashing or fast tracking.

Use the supplied mock data

Copy only the rows needed for the current step and tell the model the data is fictional. If the model changes an identifier or number, reject the change and rerun with an explicit preserve-values instruction.

Quality control

- Your chosen option is correctly named, crashing (if used) targets the critical path, and you can justify the cost/schedule/risk trade-off in three sentences.
- Compare the first AI draft with the final human-corrected artifact and retain both as evidence.
- Label assumptions, decisions, owners and dates so another project manager can audit the result.

Troubleshooting

If the response is generic, add the project objective, constraints, audience and required columns. If arithmetic is inconsistent, request formulas and substitutions, then calculate independently. If a source cannot be opened, remove the claim or replace it with verified evidence.

Submission evidence

Submit the completed project artifact, the final prompt, a short critique of the AI output, and any screenshot or exported file required by the activity.